

Transport Layer Security *(SSL and TLS)*

TCP Basics: Port Numbers

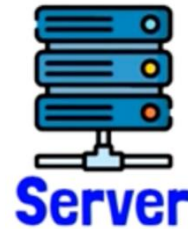
Port numbers are positive integer values in the range 0 - 65535 (16 bits), that are assigned by the Internet Assigned Numbers Authority (IANA), which are divided into 3 categories as follows :



Port number	category
0 → 1023	System ports

well-known ports

(80, 443, 21, 25)



Port number	category
1024 → 49151	user ports registered ports



ORACLE

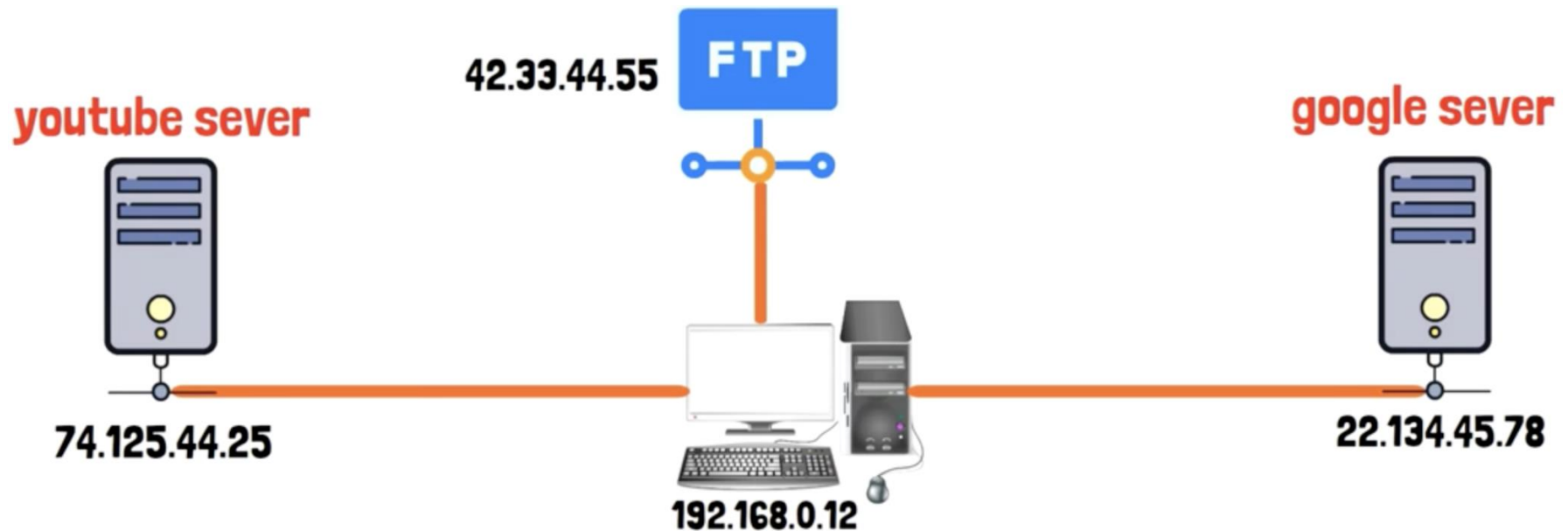


Port number	category
49152 → 65535	dynamic or private Ports



192.168.0.12 : 49536

Example: Multi Server Connections

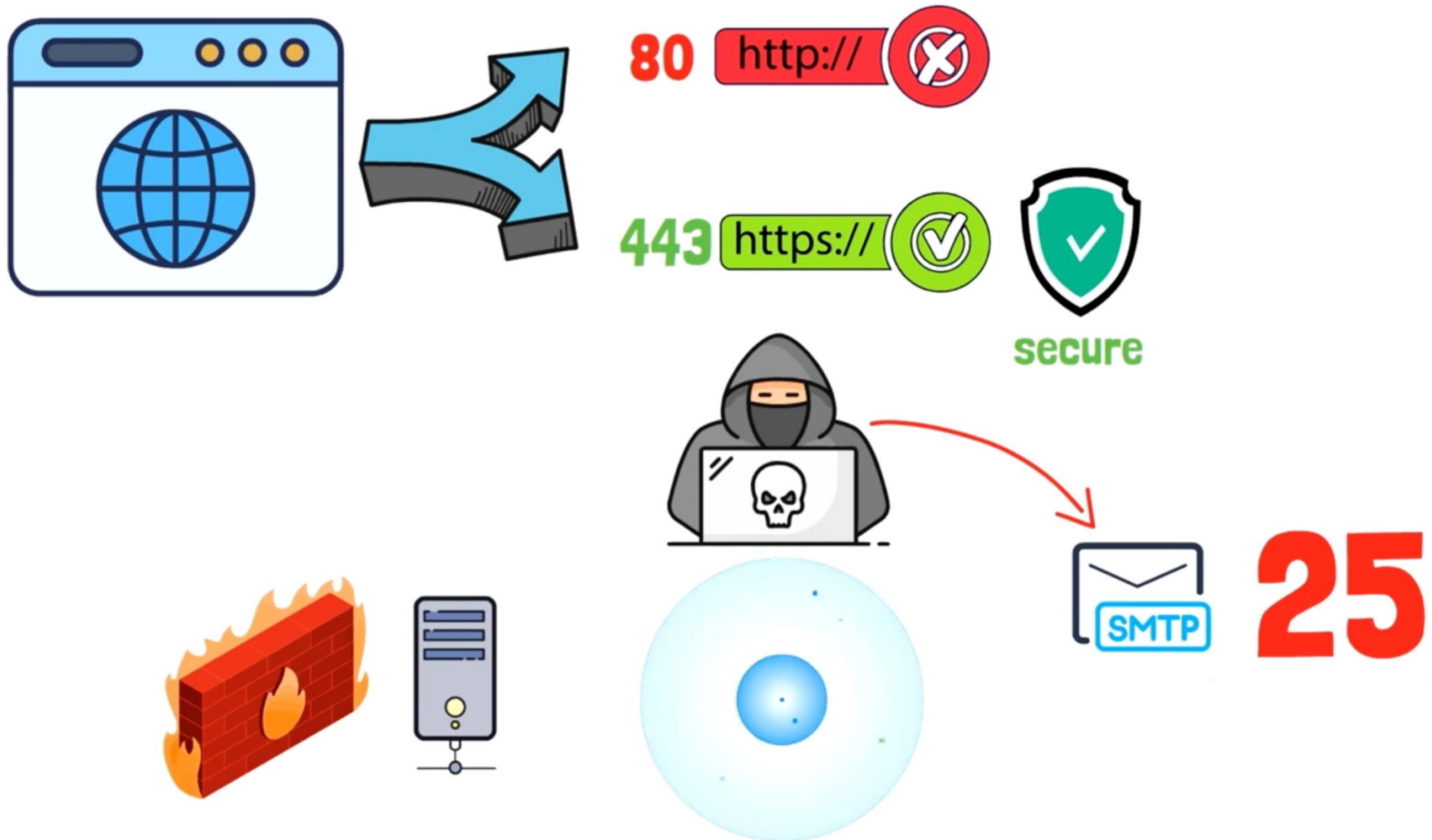


```
Command Prompt
Microsoft Windows
C:\Users\Admin> netstat -n

Active Connections

Proto  Local Address          Foreign Address         State
TCP    192.168.0.12 : 52913                74.125.44.25 : 80      ESTABLISHED
TCP    192.168.0.12 : 62976                22.134.45.78 : 80      ESTABLISHED
TCP    192.168.0.12 : 63510                42.33.44.55 : 21      ESTABLISHED
```

Example: Serving the Web Securely

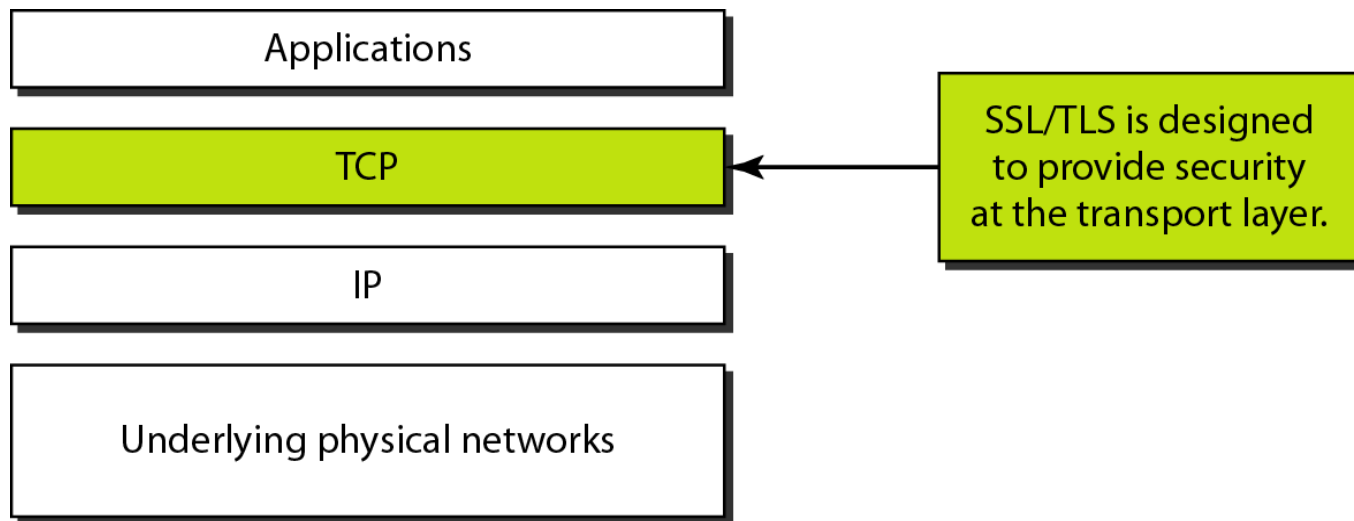


Well-Known Ports Used by TCP

Port	Protocol	Description
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

SSL/TLS

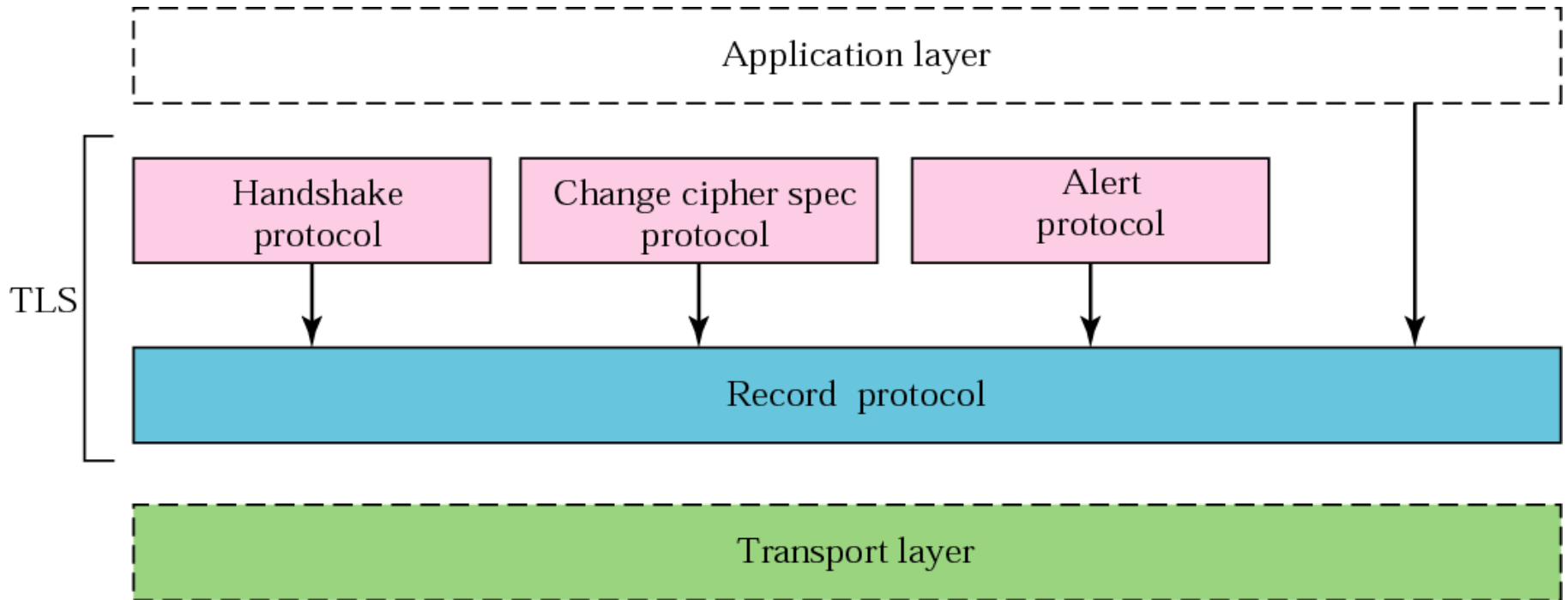
Two protocols are dominant today for providing security at the transport layer: the Secure Sockets Layer (SSL) Protocol and the Transport Layer Security (TLS) Protocol. The later is actually an IETF version of the former (designed by Netscape).



SSL (Secure Socket Layer)

- **Transport layer security service,**
- **Originally developed by Netscape,**
- **Version 3 designed with public input,**
- **Subsequently became Internet standard known as TLS (Transport Layer Security),**
- **Uses TCP to provide a reliable end-to-end service,**
- **TLS has two layers of protocols**

TLS Architecture



TLS Architecture

➤ TLS connection

- a transient, peer-to-peer, communication link,
- associated with 1 TLS session

➤ TLS session

- an association between client & server,
- created by the Handshake Protocol,
- define a set of cryptographic parameters,
- may be shared by multiple SSL connections

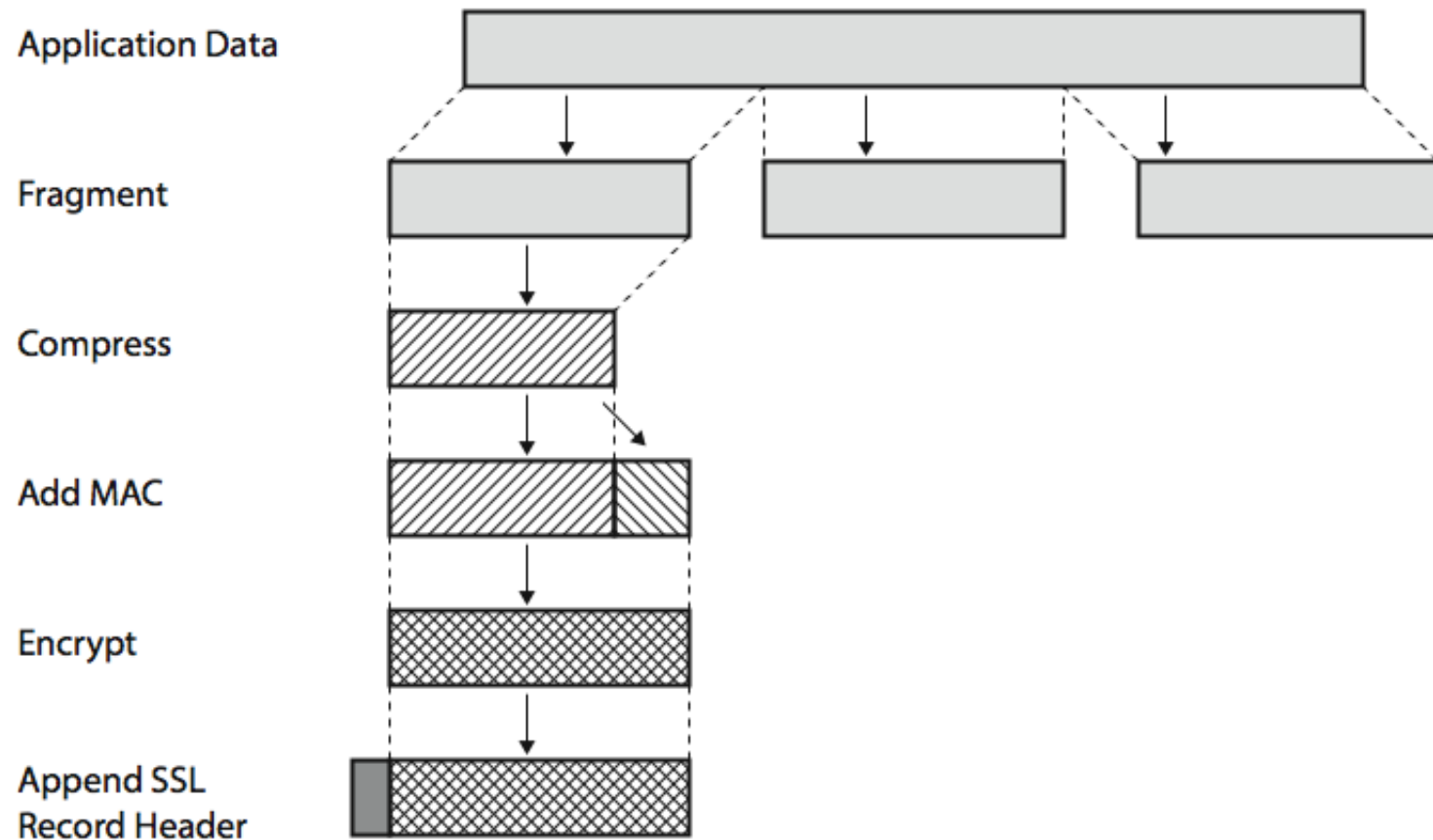
➤ Confidentiality

- using symmetric encryption with a shared secret key defined by Handshake Protocol,
- AES, IDEA, RC2-40, DES-40, DES, 3DES, Fortezza, RC4-40, RC4-128,
- message is compressed before encryption

➤ Message Integrity

- using a MAC with shared secret key,
- similar to HMAC but with different padding

Operation



HTTPS

- **HTTPS (HTTP over SSL)**
 - Combination of HTTP & SSL/TLS to secure communications between browser & server
 - documented in RFC2818
 - no fundamental change using either SSL or TLS
- **Use https:// URL rather than http://**
 - and port 443 rather than 80
- **Encrypts**
 - URL, document contents, form data, cookies, HTTP headers

Application Layer Security *(E-Mail Security)*

E-Mail Security

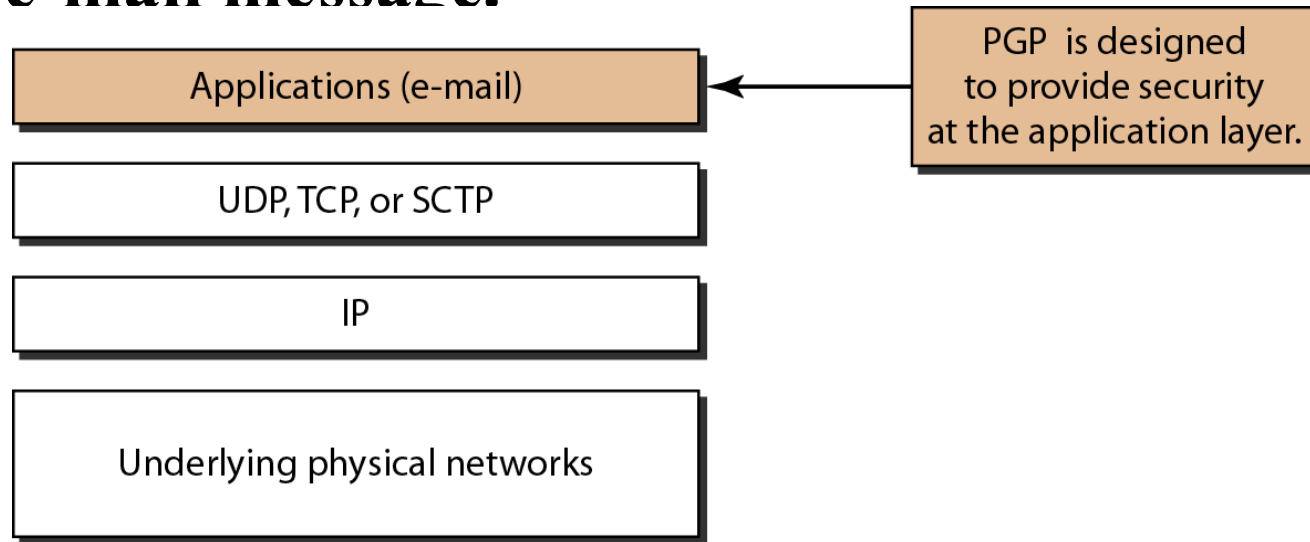
- Email is considered as one of the most commonly used network services,
- By default, the Email message contents are not secured,
- The Email contents can easily be seen either in transit or by privileged users on the destination network

E-Mail Security Enhancement

- **Confidentiality**
 - protection from disclosure
- **Authentication**
 - of the sender of a message
- **Message Integrity**
 - protection from modification
- **Non-Repudiation of origin**
 - protection from denial by sender or receiver

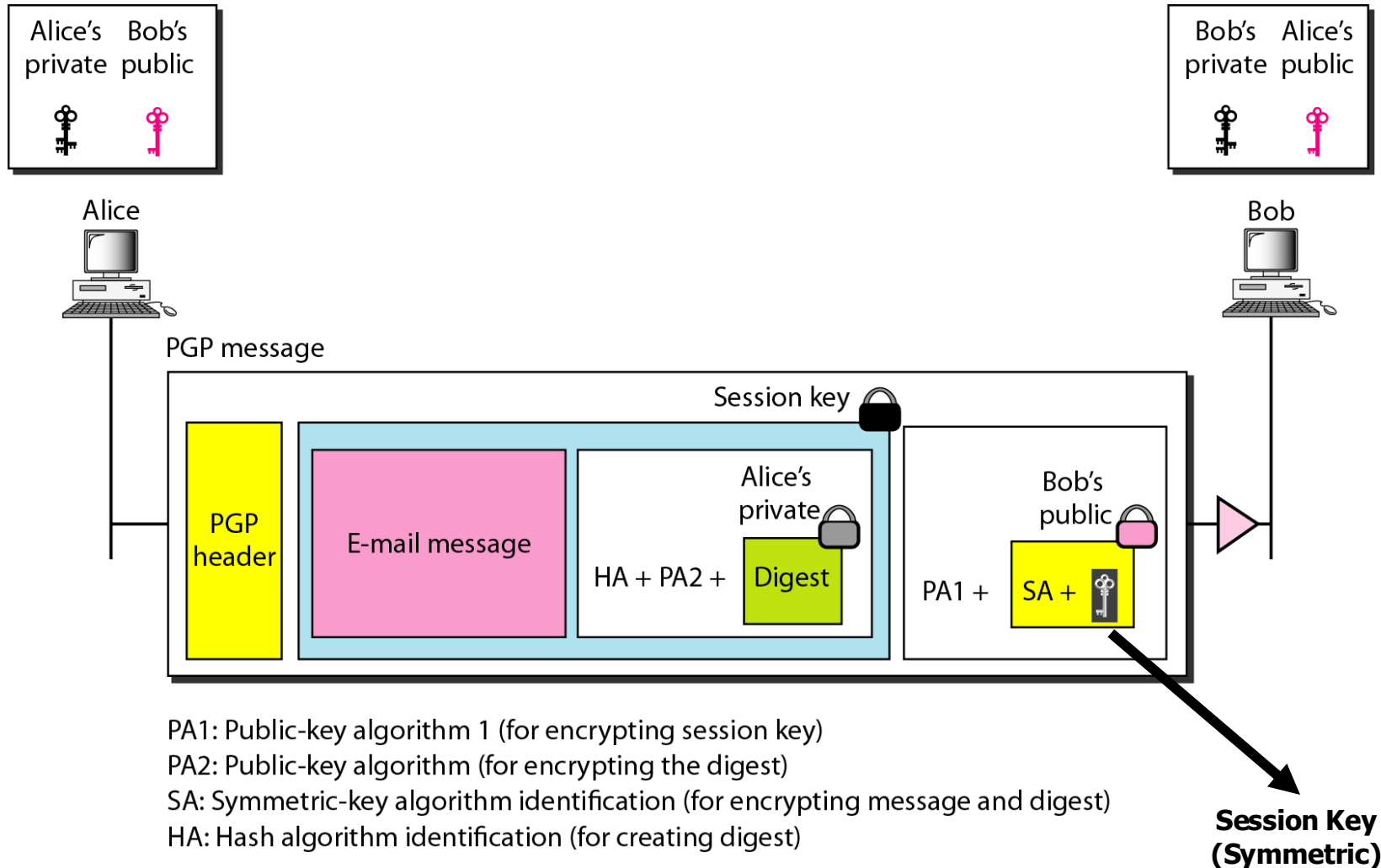
Pretty Good Privacy (PGP)

One common protocol to secure Email messages at the application layer is the Pretty Good Privacy (PGP). PGP authenticates and protects the contents of an e-mail message.



In PGP, the sender of the message needs to include the identifiers of the algorithms used in the message as well as the values of the keys.

E-Mail Message Authentication using PGP



PGP Algorithms

<i>Algorithm</i>	<i>ID</i>	<i>Description</i>
Public key	1	RSA (encryption or signing)
	2	RSA (for encryption only)
	3	RSA (for signing only)
	17	DSS (for signing)
Hash algorithm	1	MD5
	2	SHA-1
	3	RIPE-MD
Encryption	0	No encryption
	1	IDEA
	2	Triple DES
	9	AES

GPG (GNU Privacy Guard)

- GPG is a free, open-source implementation of the OpenPGP standard used for secure communication, data encryption, and digital signatures,
- It utilizes public-key cryptography (asymmetric) for key management and symmetric cryptography for fast file encryption, allowing users to encrypt files, secure emails, and verify data integrity.

GPG (GNU Privacy Guard)

- **Key aspects of GPG in cryptography include:**
 - Asymmetric Encryption: Users generate a key pair, a public key (shared) for encryption and a private key (secret) for decryption,
 - Digital Signatures: GPG allows signing documents to verify authenticity and integrity, ensuring the sender's identity and the message integrity,
 - Hybrid System: GPG combines symmetric encryption (fast) and asymmetric encryption (secure) for exchanging the symmetric key,
 - Key Management: It includes a versatile key management system to manage pairs of keys, allowing users to trust and verify others' public keys, often referred to as a "web of trust",
 - Usage: It is commonly used via command-line (Linux/macOS/Windows) or GUI tools like Gpg4win

شكراً
Dankie Gracias תודה
Спасибо Merci Takk
Köszönjük Terima kasih
Grazie Dziękujemy Děkojame
Ďakujeme Vielen Dank Paldies
Kiitos Täname teid 谢谢
Thank You Tak
感謝您 Obrigado Teşekkür Ederiz
Σας Ευχαριστούμ 감사합니다
ขอบคุณ
Bedankt Děkuje vám
ありがとうございます
Tack